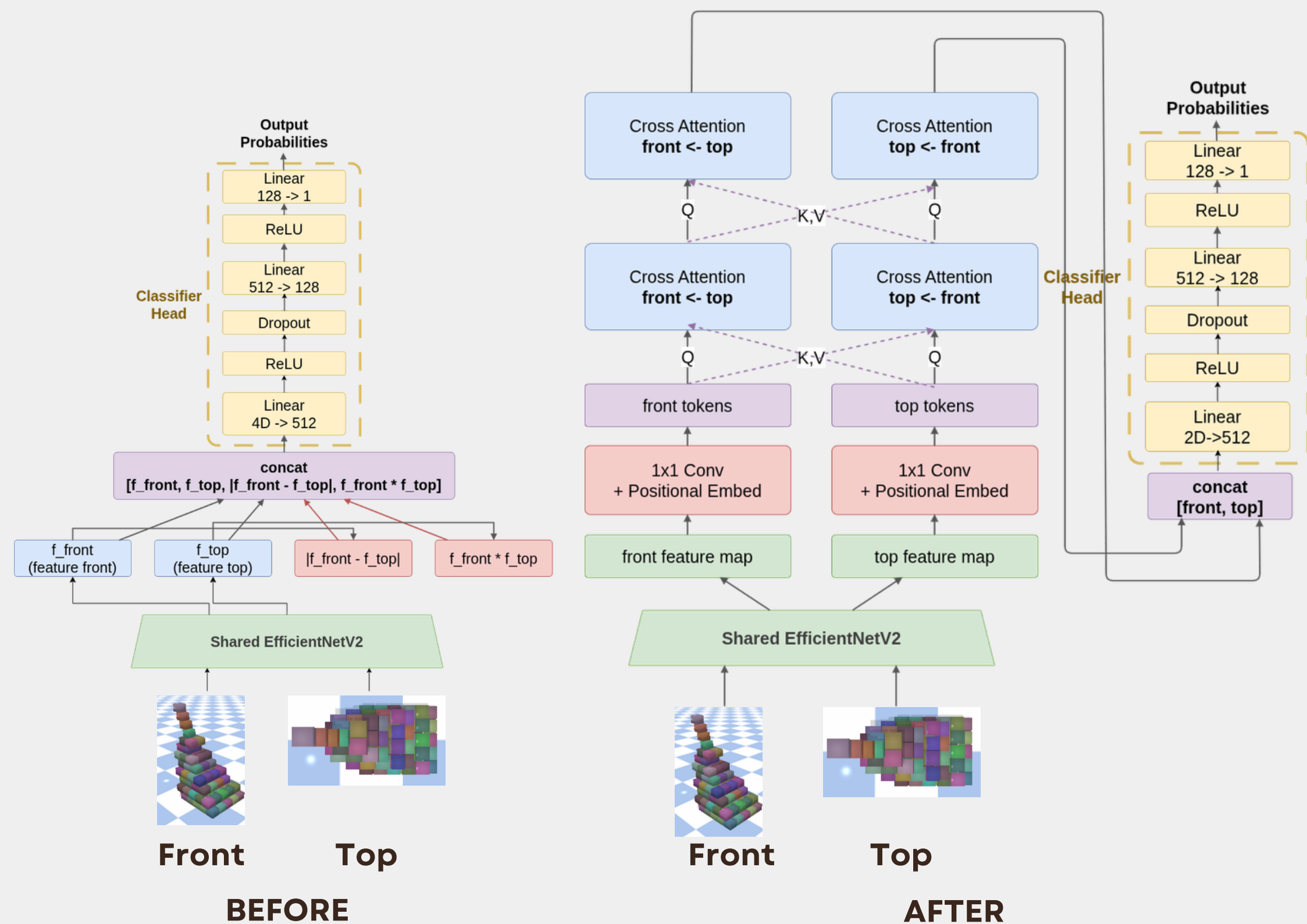


구조물 안정성 물리 추론 AI 경진대회

1. 기존 Feature Fusion 모델 대비 logloss 48.2% 및 accuracy 3.13% 개선

- Multi-View Bidirectional Cross Attention 모델 설계 및 적용



모델\val	logloss (↓)	accuracy (↑)	AUC (↑)
Feature Fusion (Baseline)	0.1216	0.96	0.9927
Cross Attention	0.2323	0.89	0.9743
Bidirectional Cross Attention (Ours)	0.063 (-48.2%)	0.99 (+3.13%)	0.9983 (+0.56%)

구조물 안정성 물리 추론 AI 경진대회

2. Baseline 대비 logloss 86.3% 및 accuracy 8% 개선

- VideoMAE로 영상 데이터에서 특징 벡터 추출 및 Distillation Regularization 추가해 학습

$$\left(\begin{array}{l} z_i^{\text{video}} = \text{VideoMAE}(V_i)_{\text{CLS}} \in \mathbb{R}^{768} \\ z_i^{\text{img}} = W_{\text{distill}} h_i + b_{\text{distill}} \in \mathbb{R}^{768} \\ \mathcal{L}_{\text{video_reg}} = \frac{1}{|\mathcal{M}| \cdot 768} \sum_{i \in \mathcal{M}} \|z_i^{\text{img}} - z_i^{\text{video}}\|_2^2 \\ \mathcal{L}_{\text{total}} = \mathcal{L}_{\text{BCE}} + \lambda \mathcal{L}_{\text{video_reg}} \end{array} \right)$$

모델	Metrics (↓)	Epochs (↓)	logloss (↓)	accuracy (↑)
efficientnet v2_rw_s (Baseline)	-	100	0.218147	0.91
efficientnet v2_rw_s (BEST)	Distillation	30	0.02986 (-86.3%)	1.00 (+9%)
efficientnet v2_rw_s	Distillation	50	0.09056	0.98
efficientnet v2_rw_s	Distillation	100	0.05361	0.99

3. Worst Trial 대비 logloss 91.2% 및 accuracy 6% 개선

- Backbone, Epochs, Batch Size 등 HyperParameter Search 수행

※ CosineAnnealingLR 스케줄러 사용, Epochs에 따라서 수렴 속도가 달라짐

순위	모델	Epochs (↓)	logloss (↓)	accuracy (↑)
1	efficientnet v2_rw_s (BEST)	30	0.02986 (-95.5%)	1.00 (+40%)
2	efficientnet v2_rw_s	50	0.090561	0.98
3	convnext tiny	30	0.113118	0.97
4	convnext tiny	50	0.322998	0.81
5	swin tiny	30	0.222444	0.93
6	swin tiny (WORST)	50	0.657001	0.60